

Internship Project Proposal

*Intelligence-Driven Purple Team Pipeline for E-Banking
Infrastructure*

Submitted By:

Mr. THY DAYUTH

Faculty of Digital Engineering
Department of Telecommunications and Networking
Telecommunication and Network Engineering

INTERNSHIP PROJECT PROPOSAL

Intelligence-Driven Purple Team Pipeline for E-Banking Infrastructure

Submitted to: **Prof. Rath**

Prepared By: THY DAYUTH

Year 3 Cybersecurity Student

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Project Title	Intelligence-Driven Purple Team Pipeline for E-Banking Infrastructure
Student	THY DAYUTH
Supervisor	Professor Rath (Cryptography / Red & Blue Team Specialist)
Host Organisation	ACLEDA Bank PLC — Cybersecurity Division
Duration	Approximately 3 Months (Internship Period)
Classification	Confidential — Internal Review Only

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1. The Problem & Objective

 THE PROBLEM Traditional SOC's are reactive and rely on generic rules. Persistent breaches stem from a lack of tested, high-fidelity detections for specific financial threats.	 THE OBJECTIVE Build a proactive Purple Team pipeline that bridges the gap between offensive emulation and defensive detection engineering.	 THE SCOPE One attack vector only: <i>Spear-phishing → Credential Harvesting → Lateral Movement → DB Access</i>
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2. Four-Phase Methodology

This project aligns with **100% of the internship curriculum** and is executed in four phases:

	Phase	What I'll Do	Key Tools / Frameworks
1	Threat Intelligence (Recon)	Profile a real financial threat group and map their TTPs using the MITRE ATT&CK framework.	<i>ATT&CK Navigator, MISP</i>
2	Red Team Emulation (Attack)	Safely execute the mapped attack chain in an isolated lab to test whether existing defences hold.	<i>Atomic Red Team, Caldera, Metasploit</i>
3	Detection Engineering (Defence)	Analyse generated logs to author and tune high-fidelity, low-noise SIEM detection rules.	<i>Sysmon, SIEM (Splunk / ELK), Sigma</i>
4	IR Playbook (Response)	Document a Tier 2/3 containment and eradication playbook aligned with BCP/DRP and risk frameworks.	<i>NIST IR Framework, BCP/DRP</i>

3. Final Deliverables

At the end of the internship, three immediately usable security artefacts will be delivered:

01 EMULATION RUNBOOK A repeatable procedure documenting the full attack path and tools used.	02 SIEM DETECTION RULES 3+ tested alert rules (Sigma / native query) — ready for deployment at <5% false-positive rate.	03 IR PLAYBOOK A step-by-step Tier 2/3 response guide tailored to the emulated e-banking threat.
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